

NATIONAL UNIVERSITY OF MODERN LANGUAGES

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System ID: NUML-S22-15676
Department: Software engineering
Section: BSSE-1-A- Afternoon
Submission date: 12-03-2022

Q1 What are the typical c++ environment phases of a c++ program? Explain each phase in your own words.

Typical c++ Development Environment:

- * C++ system generally consist of three parts:
 - ⇒ A program development environment.
 - ⇒ The language
 - ⇒ The c++ standard library.
- * C++ programs typically go through six phases:
 - ⇒ edit ⇒ preprocess ⇒ compile
 - ⇒ link ⇒ load ⇒ execute

Phase # 1:

Phase 1 consists of editing a file with an editor program, normally known simply as an editor.

- * Type a c++ program using the editor
- * Make any necessary corrections.
- * save the program.
- * C++ source code filenames often end with # the .cpp, .cxx, .c extensions, which indicate that a file contains c++ source code.

Integrated development environment (IDEs)

it provide tools that support the software-development process, including editors for writing and editing programs.

for example

- ⇒ Visual studio
- ⇒ Dev c++ etc

Phase # 2:

- * In phase 2, you give the command to compile the program.
- A preprocessor program executes automatically before the compiler's translation phase begins.
- The C++ preprocessor program executes automatically before the compiler's translation phase begins.
- The C++ preprocessor obeys commands called preprocessing directives, which indicate that certain manipulations are to be performed on the program before compilation.
- These manipulations usually include (copy into the program file) other text files to be compiled, and perform various text replacements.

Phase # 3:

- * In phase 3, the compiler translates the C++ program into machine-language code - also referred to as object code.

Phase # 4:

Phase 4 is called linking.

- The object code produced by the C++ compiler typically contains "holes" due to these missing parts.
- A linker links the object code with the code for the missing functions to produce an executable program.
- If it compiles and links correctly, an

executable image is produced.

Phase # 5:

Phase 5 is called loading.

- Before a program can be executed, it must first be placed in memory.
- This is done by the loader, which takes the executable image from disk and transfers it to memory.
- Additional components from shared libraries that support the program are also loaded.

Phase # 6:

Phase 6 is called execution:

- Finally, the computer, under the control of its CPU, executes the program one instruction at a time.
- Some modern computer architectures often execute several instructions in parallel.

The programs might not work due to various errors. If this happened then we have to return to the edit phase and make the necessary corrections and proceed again.

Q2 What are different Escape Sequences in C++?

Escape Sequences:

Escape sequence is a set of characters that convey a special meaning to the program.

They are used to convey a meaning

which cannot be conveyed directly. Escape sequence and their functions are given below.

\a:

Alert sound. A beep is generated by the computer on execution.

\b:

It moves the cursor one space back

\f:

Form feed, this escape sequence is used in printing. It is used to insert a blank page in the printed output.

\n:

New line, shifts the cursor to new line. Words on the right of "\n" go to next line.

\r:

Carriage return. Positions the cursor to the beginning of current line.

\t:

Horizontal tab. It moves the cursor by a number of spaces or to next tab stop.

\v:

vertical tab.

\':

Displays a single-quote character (').

\":

Displays double-quote character (").

\?:

Displays question mark (?).

\0:

Null character, marks the end of string.

Q#3 Explain the use of setw() function of iomanip header file using coding examples?

The setw manipulator is used in output statement to set the minimum field width.

It has the general form.

setw(n)

- Here, n is an integer value that causes the number or text that follows to be printed within a field width of n characters.
- The number or text is right justified within the set field width.
- It is commonly used in program to align number or text on output.

The following program demonstrate the use of setw manipulator in a program.

```
#include <iostream>
```

```
#include <iomanip>
```

```
main()
```

```
{
```

```
    int price1 = 500, price2 = 1000, price3 = 50000;
```

```
    cout << "product" << setw(10) << "price" << endl;
```

```
    cout << "Book" << setw(10) << "Price" << endl;
```

```
    cout << "bag" << setw(10) << "Price" << endl;
```

```
    cout << "computer" << setw(10) << "Price" << endl;
```

```
}
```

Out put :-

Product	Price
Book	Price 500
bag	Price 1000
computer	Price 50000

* Without the header file 'iomanip', this type of program cannot run where setw() manipulator are used in the program.

* iomanip.h

iomanip.h provides manipulator function such as setw(), setprecision(), etc

THE
END